Mini-Lesson 2: Collisions Pre-Lab

AP Physics 1 · Unit 4 · Do with the mini-lesson before Lab 4.1 on Fri 11/13. **Bring this sheet to lab.** Right is + on this whole sheet.

Scan with your iPad camera to open the lesson

Name _____ Period _____

A Three kinds of interactions

BEFORE | AFTER

Bounce



sketch after lab

Stick



sketch after lab

Explode



sketch after lab

A **collision** starts _____. The carts either bounce apart or _____.

An **explosion** starts _____ and ends _____.

B Two carts, one total

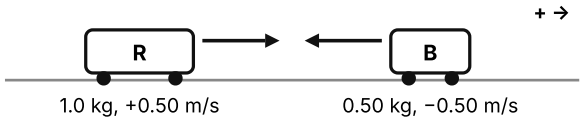
p_{total} = _____ + _____

Add momenta with their _____.

Carts moving toward each other partly _____, so the total can be smaller than either one, or even _____.

From the slider: I made the total zero with red = _____ kg at _____ m/s and blue = _____ kg at _____ m/s.

C Worked example: one row of the lab table (Trial 4, head-on)

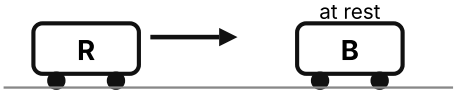


Trial	Initial (kg·m/s)			Final (kg·m/s), tomorrow		
	p red	p blue	p total	p red	p blue	p total
4	_____	_____	_____			

The speeds are the same, but the total points _____ because the _____ cart has more momentum.

D Predictions: fill in "Lab result" during the lab

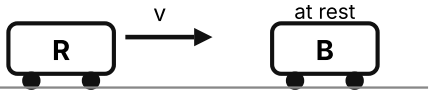
1. Bounce: identical carts



My prediction: _____

Lab result: _____

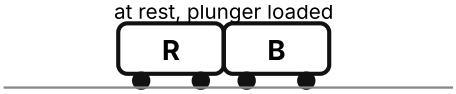
2. Stick: identical carts



My prediction: _____

Lab result: _____

3. Explode: identical carts



My prediction: _____

Lab result: _____

Predictions aren't graded. Write what you think, then check it in lab.

E Check questions: show your work

1. 0.75 kg blue cart, left at 0.40 m/s

work:

p = _____

2. Red 0.50 kg right 0.80 m/s; blue 1.00 kg left 0.30 m/s

work:

p_{total} = _____

3. Explosion carts at rest: initial p_{total} = _____, because _____

Tomorrow's big question: When carts collide or explode, what happens to p_{total}? My guess:

_____

After the lab: _____